

DeRexi Core System Components



Dana
Policy Owner -
Compliance
Department



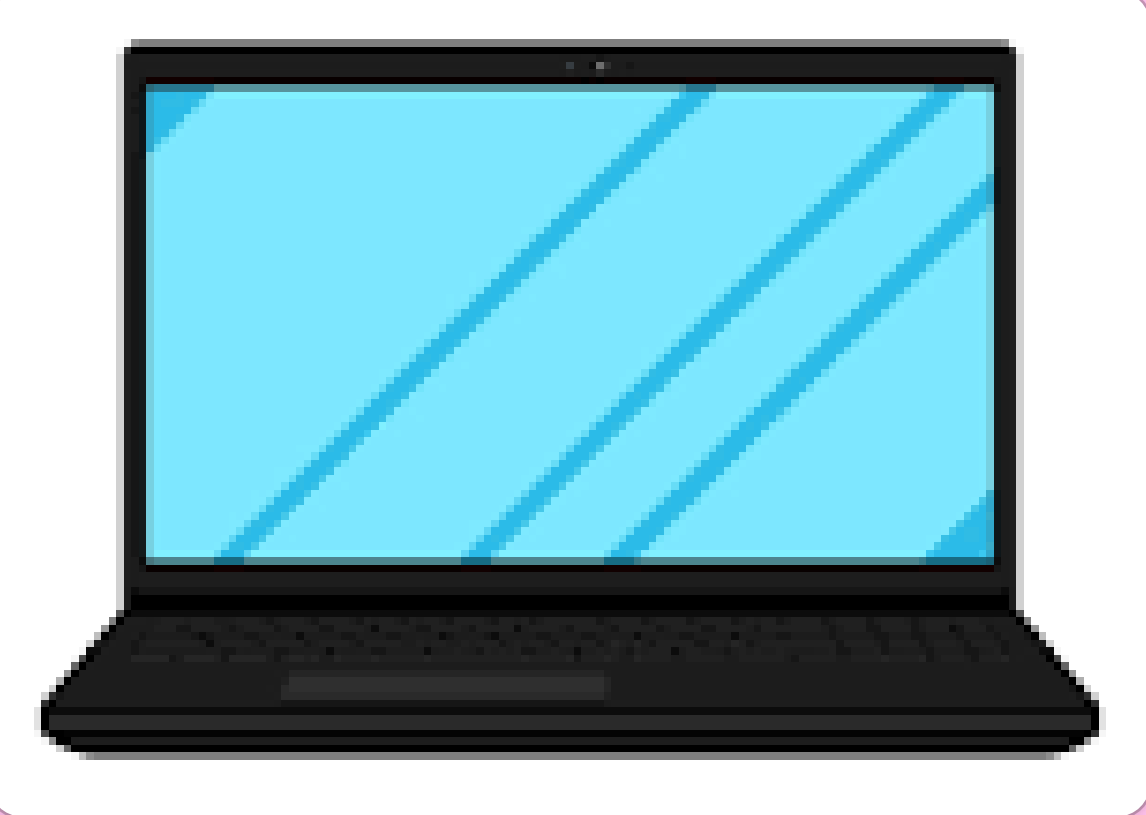
Lucy
DeRexi System
Administrator



Tom
New Employee
Onboarding -
Operations



Maya
Marketing Associate



Example Device:
Dana's Work Laptop

It is important that the **TCP method** is used at the Transport layer because Dana is accessing company policies and potentially updating important information. We want the information to arrive completely and correctly, not simply as fast as possible.

OSI Model	What it means for Dana's laptop	TCP/IP Model
7. Application	Dana uses a browser to open DeRexi	Application
6. Presentation	Data is formatted and encrypted, such as with TLS	Application
5. Session	Maintains Dana's active login/session	Application
4. Transport	TCP reliably delivers the data	Transport
3. Network	Dana's laptop uses an IP address and packets are routed	Internet
2. Data Link	Her laptop uses its MAC address to communicate on the local network	Network Access
1. Physical	Wi-Fi or Ethernet physically carries the signal	Network Access

Setting	Example	What Does This Mean
Network	192.168.10.0/24	192.168.10.0 represents the subnet itself.
Subnet Mask	255.255.255.0	Defines which portion of the IP identifies the network and which portion can identify individual devices.
Dana's Laptop IP	192.168.10.25	This is Dana's individual address inside that network.
Default Gateway	192.168.10.1	This is the door out of the subnet. If Dana needs to communicate with something outside her local network, her traffic goes toward this gateway.
Usable device range	192.168.10.1 – 192.168.10.254	IP addresses within this range are assigned to other devices under the Compliance department's network subnet.
Broadcast Address	192.168.10.255	This is a special address used to communicate with all devices within Dana's subnet.

